



Final Exam, Theory of Computation 2024

1 (15 pts) **Quick-fire round.** Consider the following statements.

1. Every language in $\mathbf{P}$ is regular.
2. Every finite language is regular.
3. Every language in $\mathbf{NP}$ is decidable.
4. Language $\{0^n 1^m : n \leq m\}$ is regular.
5. Language $\{w : \text{number of 1s in } w \text{ is divisible by } 2024\}$ is regular.
6. Language $\{\langle D \rangle : D \text{ is a DFA with } L(D) = \emptyset\}$ is in $\mathbf{P}$.
7. Language $\{\langle G \rangle : G \text{ is a graph with vertex cover of size } 2024\}$ is $\mathbf{NP}$ -complete.
8. Suppose $A \subseteq B \subseteq C$ where A and C are decidable. Then B is recognisable.
9. If A and B are recognisable, then $A \cap \overline{B}$ is recognisable.
10. If $A \leq_m B$, then $A \leq_p B$.
11. If $A \leq_m B$, then $\overline{A} \leq_m \overline{B}$.
12. If $L \in \mathbf{NP}$, then $\overline{L} \in \mathbf{NP}$.
13. $\mathbf{P} = \mathbf{NP} \cap \mathbf{coNP}$.
14. $3\text{-SAT} \leq_m \text{HALT}$.
15. $2\text{-SAT} \in \mathbf{P}$.

For each statement, write down one of

- **T** if the statement is known to be true.
- **F** if the statement is false *or not known to be true*. E.g., both $\mathbf{P} = \mathbf{NP}$ and $\mathbf{P} \neq \mathbf{NP}$ should be marked **F**.
- or leave your answer empty.

A correct **T/F** answer is worth +1 point, an incorrect answer is worth −1 point, and an empty answer is worth 0 points.

2 (20 pts) Regular languages.

2a Write down the statement of the Pumping Lemma. (*Only the statement, not its proof!*)

2b Let $\Sigma = \{a, b, c\}$. Show that the following language is regular:

$$\text{DOUBLE-}\Sigma = \{w \in \Sigma^* : \text{Every symbol in } \Sigma \text{ appears in } w \text{ at least twice}\}.$$

3 (15 pts) NP-completeness I. Show that the following language is **NP**-complete:

$$\text{DOUBLESAT} = \{\langle \varphi \rangle : \varphi \text{ is a CNF that has at least two satisfying assignments}\}.$$

4 (15 pts) NP-completeness II. Suppose $\mathcal{S} = \{S_1, \dots, S_n\}$ is a family of subsets of $\{1, 2, \dots, n\}$. We say that k distinct sets $S_{i_1}, \dots, S_{i_k} \in \mathcal{S}$ form a *packing of size k* if the sets are pairwise disjoint, that is, $S_{i_j} \cap S_{i_{j'}} = \emptyset$ for $j \neq j'$. Show that the following problem is **NP**-complete:

$$\text{SETPACKING} = \{\langle \mathcal{S}, k \rangle : \text{Family } \mathcal{S} \text{ contains a packing of size } k\}.$$

5 (20 pts) Undecidability. Consider the language

$$\text{DOUBLETM} = \{\langle M \rangle : M \text{ is a TM that accepts at least two inputs, i.e., } |L(M)| \geq 2\}.$$

Classify DOUBLETM as one of (i) decidable, (ii) undecidable but recognisable, (iii) unrecognisable. Justify your answer with a proof. (*Note: You are not allowed to use Rice's theorem as mentioned in Homework 2! Use only concepts introduced in class/exercises.*)

6 (15 pts) Doubly accepting NFA. Let N be an NFA. The usual definition of acceptance is the existence of at least one accepting path. Change this definition as follows: N accepts the input word w iff there are at least two accepting paths. Denote by $L_2(N)$ the set of strings that admit at least two accepting paths. Prove that $L_2(N)$ is regular.